Lab 2 Research questions

1. **Do learned features beat hand-crafted descriptors?** Compare all models on the same 200-song test set. You will need to rerun the Lab 1 models on the new split for a fair comparison.
2. **Do CNNs and descriptor models make the same mistakes?** Compare your confusion matrices and check how much their errors overlap. Is the genre-similarity structure found in Lab 1 the same here, or is it partly an effect of the descriptors?
3. **Are the two approaches complementary?** Concatenate CNN embeddings with the Lab 1 descriptors, or ensemble the two models. If their errors don't overlap much, the combination should improve accuracy.
4. **Is the extra cost worth it?** Compare accuracy against training time, inference time and model size. A random forest on descriptors may come close to a CNN at a fraction of the cost.
5. **How much does one 80/20 split tell you?** With 200 test songs, one song is 0.5% of accuracy. Repeat the split with several seeds, before claiming that one model beats another.
6. **How does data size affect the comparison?** Plot learning curves using 10%, 25%, 50% and 100% of the training set for the scratch CNN, and the best Lab 1 model.

7. **Should a CNN treat a spectrogram like a photo?** Compare square 3×3 kernels with "musically motivated" shapes (as in Pons et al.).
8. **How do depth and parameter count impact?** Plot test accuracy against the number of parameters (in different CNN configurations tried). Where does overfitting start?
9. **Would fine-tuning a pretrained model (e.g. VGG16, VGGish Clap) improve the model?** Is a vision model or an audio model the better starting point?
10. **Frozen embeddings + traditional ML algorithm, or full fine-tuning?**